NLP project: Topic Modeling & Temporal Evolution The place of ecology as a political subject between 1973 & 1993

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Abstract

In this study, we employ NLP tools like the Latent Dirichlet Association model and the BERTopic model to investigate the place of ecology as a theme among the political landscape between 1973 and 1993, using the political manifesto published for legislative campaigns for this period.

Disclaimer: part of this work as been realised with the cooperation of fellow student Mathis Lecoq. We worked together to adapt the dataset and select our models, however we employed it to different ends. I would also like to use this opportunity to thank him, for without his contributions this study couldn't have been fully realised.

1 Introduction and literature review

Automated analysis of political texts belongs to a broader literature on "text as data" methods, which aim to quantitatively measure the priorities, positions, and discursive strategies of political actors (Gentzkow et al., 2019). Electoral programs are a key source for this type of analysis, as illustrated by the Manifesto Project, which manually codes party manifestos into thematic categories in order to measure partisan priorities (Lehmann et al., 2025). This project led to several studies, like how citizens withdraw their support for political parties that have failed to turn their pre-election pledges into tangible policies (Matthieß, 2024).

This project relies on a corpus of political manifestos from the French legislative elections between 1973 and 1993, excluding the 1986 election (Bougon et al., 2023). The dataset contains the manifesto of each candidate in each constituency, offering a rich source of candidate-level political communication. Unlike national party platforms, these texts capture both national campaign themes and more localized concerns. Furthermore, candidates may also react to their opponents in their manifestos, which is probably less prevalent in party manifestos. On the contrary they can avoid it and focus on their own thematics to have *issue ownership* (Stubager, 2018). Or even refer to the President of the Republic, a key figure in the semi-presidential system in France.

The longitudinal nature of the data allows us to analyse how political themes evolve over time and how parties adapt their discourse to changing electoral contexts. While the number of manifestos in each side allows us to really compare the priorities of each party. These points also lead to challenges: the manifestos may differ in rhetorical style, in local specificity, in lexical choices. And the long time span may lead to the appearance of new topics and the disappearance of others.

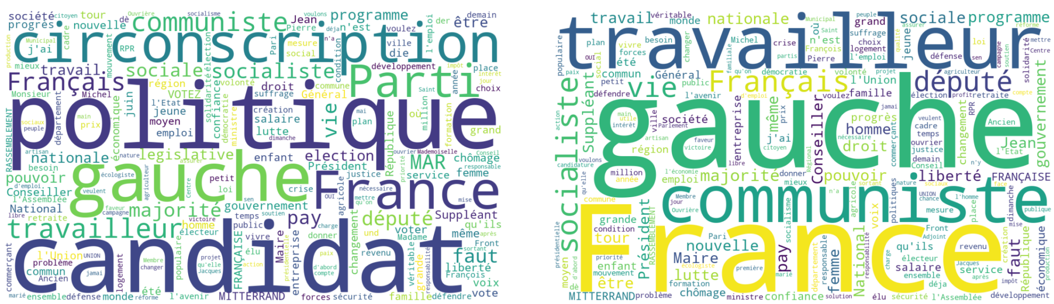
Specifically, one would expect ecology matters to rise as the years pass. In this work, we will try to examine how this subject has settled in the public debate throught the study of this corpus, across all political groups and with a focus on relevant parties.

2 Data presentation, initial data analysis

2.1 Data preparation

The corpus is composed of 21,167 manifestos spanning 5 elections, with on average 692 words per manifesto. For each text we have information about its context and the candidate. The variables are presented in annex A. As we want to evaluate topic evolution across parties, we are interested in the variables *date* and *titulaire-soutien*, which correspond to the party affiliated with the candidate. However a lot of political party are present, and they may change names across the 20 years spanned by the corpus. So we reordered them into 12 political groups: independent/micro-party, ecologist, autonomist, specific cause, far-left, left, center-left, center, center-right, liberal right, gaullist right, and far-right.

We removed some artefacts from the texts, and used a list of stop words to remove the common but unimportant words from the analysis. But we had to complete the list with words from the election context, otherwise we had to many words with no meaning in the context of our analysis as they just refer to the election context, like "politique", "candidat", or "député", as shown in figure 1.



(a) with classic stop words

(b) adding specific stop words

Figure 1: Global word-clouds on the whole dataset.

2.2 Global analysis

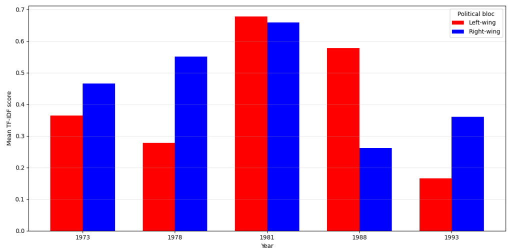
Based on analysis of the top 50 words and 50 pairs of words (which are presented in Annex B), we can observe that candidates do not only present their programs and ideologies, but also situate their discourse within a broader electoral, institutional, and social context. 5 broad categories are observed and are presented in Table 1.

Table 1: Presentation of the broad categories

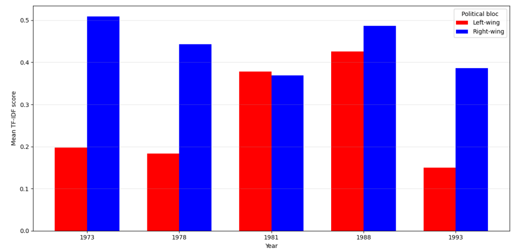
Category	Description	Associated words
`Party and coalition`	References to parties, alliances and major political actors	gauche, front national, union gauche, François Mitterrand
`Candidate legitimation`	References to close high level political figures and to current/previous mandates	François Mitterrand, Giscard d'Estaing, maire, conseiller régional
`Political program`	References to their political program	pouvoir d'achat, services publics, chômage, allocations familiales
`Social groups`	References to specific social groups and address them	femmes, enfants, petits commerçants, travailleurs
`Values, collective identity`	References to common values and identity with voters	liberté, progrès, france, changer vie

2.3 First temporal evolution

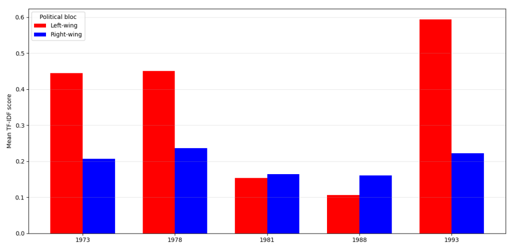
To first show that politician adapt their topics across time and elections, we first use a first analysis on the frequency of some words. We use term frequency-inverse document frequency (TF-IDF) method in order to compare the relative frequency of each word for different political groups.



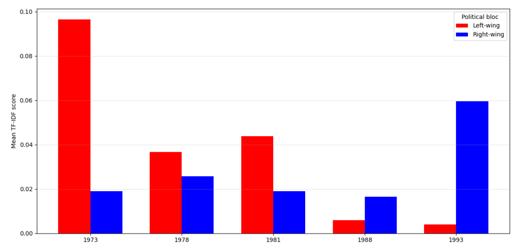
(a) majorité



(b) président



(c) travail



(d) armée

Figure 2: TF-IDF evolution of left-wing and right-wing parties for:

Figure 2 shows the evolution of 4 words for left-wing (Extrême-gauche, Gauche, Centre gauche) and right-wing (Extrême-droite, Droite, Centre droit) parties. We can clearly see a surge in the terms “majorité” and “président” in 1981 and 1988 for left-wing parties. This correspond to the legislative elections took place just after the elections of François Mitterrand, a left-wing politician, to presidency. So left-wing candidates used it as a way to legitimize themself and rally opinion. And on the contrary passed less time with programmatic propositions, as we can see with the decrease in the term “travail”. We observe the opposite dynamic for right-wing politician for the word “majorité”[1]: once again they capitalize on the political situation and focus less on programmatic propositions.

However candidates still express political opinions and are not just trying to please electors, but still express their ideology. For example, the word “président” continues to appear in 1981 and 1988 in right-wing parties. This can be explained by the greater importance the figure of the president plays on the Right compared to the Left.

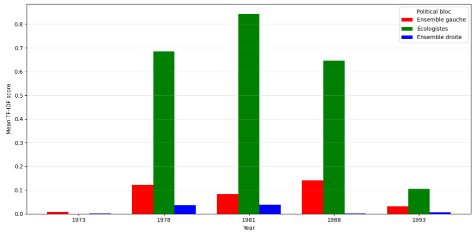
Candidates also adapt their program to new situations in the world. For example the fall of the Soviet Union in the early 90s led to program changes in right-wings parties, which appears clearly with a spike in 1993 on Figure 2d.

Finally parties also react to topics pushed by their opponents. Figure 3 shows that the apparition of ecologist parties, with a specialization on some subject, like the nuclear, forces other parties to align themself on this subject. Ecologists who are forced to talk less about this issue in 1993 and react to the creation of a focus group about hunting (*Chasse, pêche, nature et traditions* in 1989).

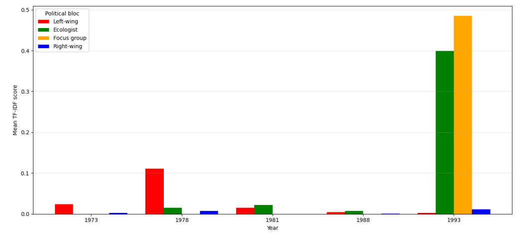
3 Models presentation

To analyse the thematic structure of the corpus, we rely on two complementary topic modelling approaches: Latent Dirichlet Allocation (LDA) (Blei et al., 2003) applied separately to political groups, and BERTopic’s Topics over Time approach to study temporal evolution (Grootendorst,

[1] Except in 1981, but this can be explained by being the first transfer of power from the Right to the Left in the fifth Republic: majoritarian logic was not fully implemented.



(a) nucléaire



(b) chasse

Figure 3: TF-IDF evolution of left-wing, right-wing, ecologist parties, and focus group for:

2022). LDA is a probabilistic unsupervised model that represents each document as a mixture of latent topics, and each topic as a distribution over words. In our analysis, estimating separate LDA models for each political group allows us to identify the main themes emphasized by each side of the political spectrum. This approach is relatively simple to implement and easy to interpret through the most probable words associated with each topic. However, it also has limitations: since models are estimated independently for each group, the resulting topics are not always directly comparable across groups, and their interpretation remains partly subjective.

To study temporal evolution, we complement LDA with BERTopic's dynamic topic modelling functionality, known as Topics over Time. BERTopic differs from traditional LDA-based models because it relies on transformer-based document embeddings, clustering, and class-based TF-IDF (c-TF-IDF) to generate interpretable topics. In the Topics over Time approach, the model is first fitted on the whole corpus without explicitly taking time into account. Then, for each topic and each time period, BERTopic recalculates the c-TF-IDF representation in order to observe how the vocabulary associated with a topic changes over time. This is particularly relevant for our corpus, which covers several legislative elections between 1973 and 1993, because it allows us to track the evolution of political themes across elections while keeping a global topic structure. Its main advantage is that it captures semantic similarity more effectively than purely word-frequency-based models and provides a practical way to analyse temporal changes in topic representations.

4 Global analysis of the ecology subject across elections

The first step is to examine how prevalent the ecology is as a subject in general. Indeed, if all political groups are tackling this issue, the difference would be minimal and this question would be rather pointless. Our first step is then to fit an LDA over all periods of time, and all political groups. We first apply a count vectorizer to get rid of all the stop-words from our corpus, remove accentuation for a more cohesive approach, and keep only the terms that are present in less than 95% of documents and more than 2 of them. Once we have our numerical representation, we can feed it to an LDA producing a maximum of 10 topics.

We visualize it by projecting the results on two dimensions and obtain the Inter-topic Distance Map 4:

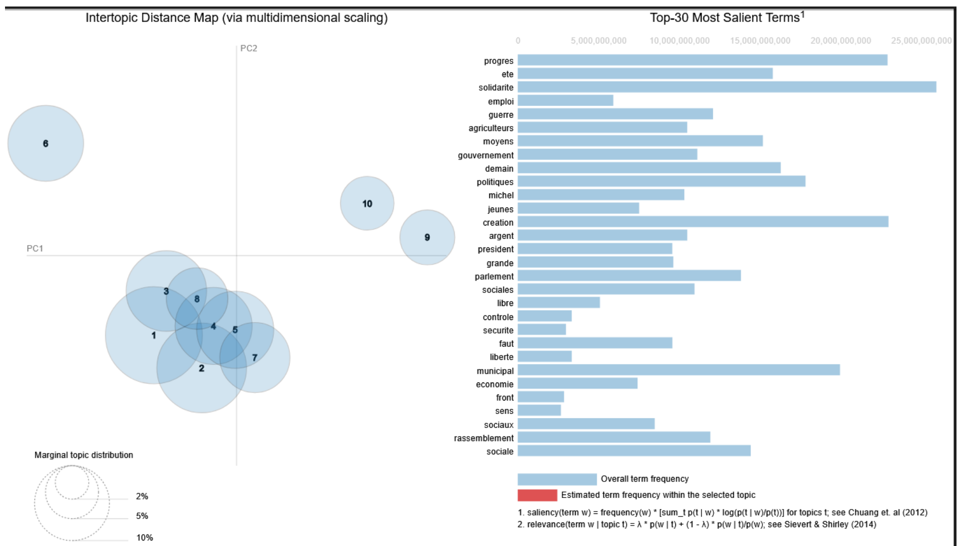


Figure 4: Intertopic Distance Map and the 30 most salient terms for an LDA fitted over the whole dataset

We can notice that none of the 30 most salient terms are related closely to the ecology subject, so it seems that it is not an ubiquitous and timeless question. After taking a closer look, we can see that the term "environment" is in the top 30 most salient words of two topics, topic 5 5 and topic 9 6. However, it is not that frequent in the selected topics, so it's still not a central question. We can thus conclude that, in general, the ecology was not a subject of its own for most political groups, but rather a matter treated on the side of broader preoccupations.

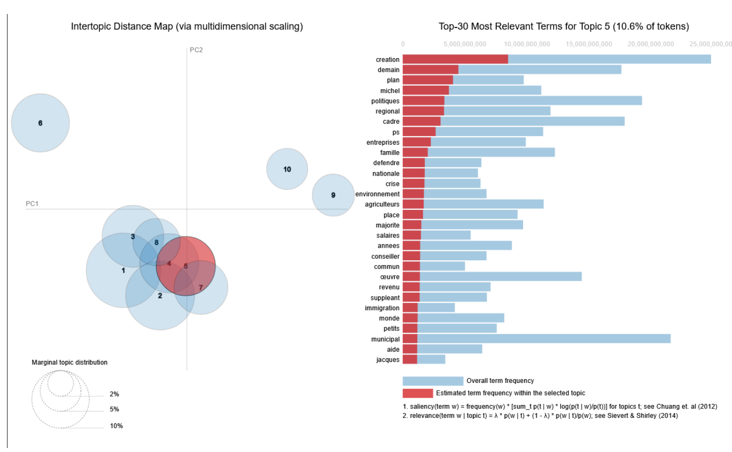


Figure 5: The 30 most salient term for topic 5

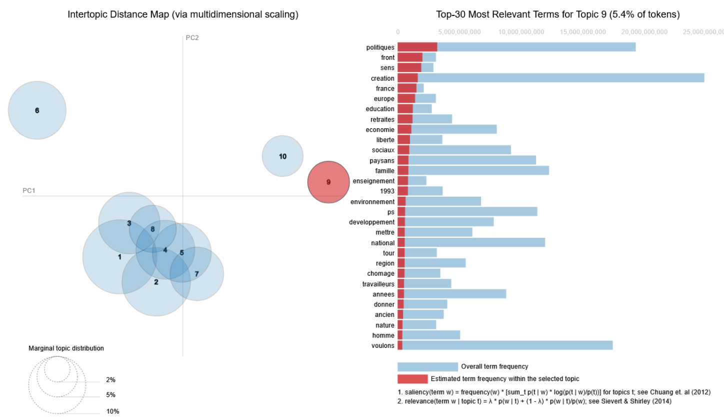


Figure 6: The 30 most salient term for topic 9

Even if it was probably not a central question for most political groups, one can guess that it still was, at least for the Ecologist parties. We can take a look at this assumption by looking at the 4 top most salient words for the 5 biggest topics in different political groups by fitting LDA on them specifically, with the same specification as before except we only generate 5 topics.

First, if we take a look at the LDA generated for the Ecologist group 7.

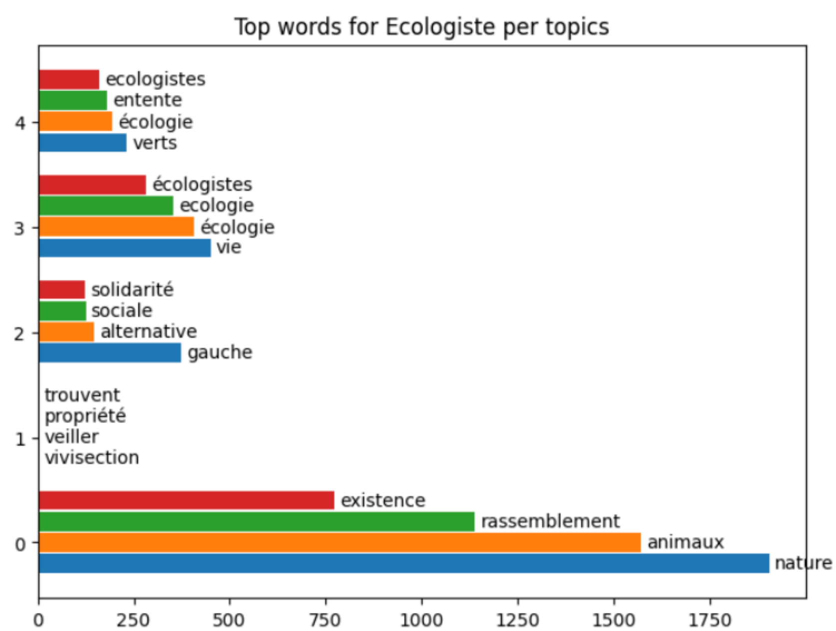


Figure 7: Most frequent words per topic for the Ecologist political group

It is to be noted that the words in topic 0 are not necessarily more frequent than the other, however it shows that the topic 0 is very focused on a few words. Here we can see that this topic is centralized on animals and nature, and shows that those party try to get into more detailed aspects of the subject. We also see that topic 2 illustrate that ecologist parties are in general very tied to left ideologies. It is thus of interest to see if there's representation of ecology related words in the LDA for the left and far-left groups.

Regarding the far-left group, we obtain the top words illustrated in 8. We can denote that just as for the global analysis, ecology was not always an issue tackled by far-left parties, as none of the corresponding vocabulary figures among the top topics for this group. We also notice that we tried to

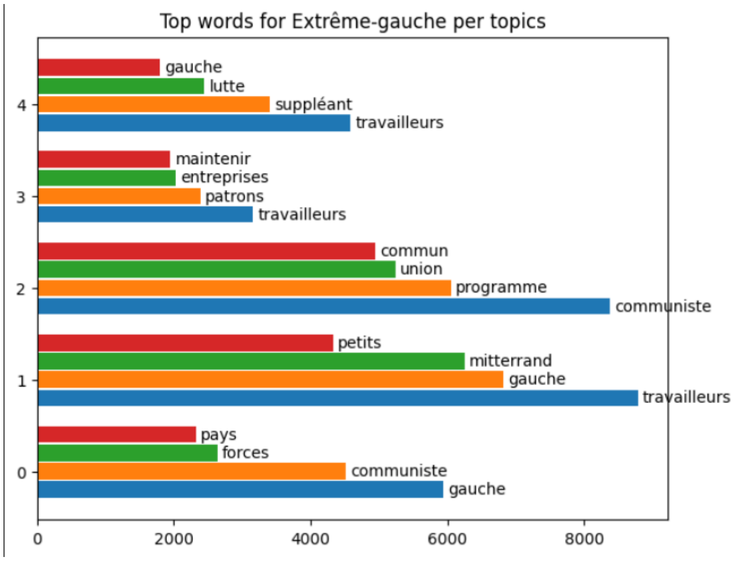


Figure 8: Most frequent words per topic for the Far-Left political group

push the number of topic to 10 for the more moderate left parties, but ecology matters didn't appear either.

Those considerations are not surprising, since the rise of ecology as a political argument is perceived as a very recent phenomenon. We thus want to take a look at the evolution of those topics over time.

5 Dynamic Topic Modelling

In this section, we will forsake LDA models, as those are static models and poorly suited for temporal and dynamic modelling. We instead use BERTopic methods, that employ transformers and vectorizers to cluster the words into topic. The model's pipeline in the context of dynamic topic modelling is illustrated in 9, the main take away being that we first fit the model over the data as if it was timeless to create the topic, and then evaluate how it evolves over time.

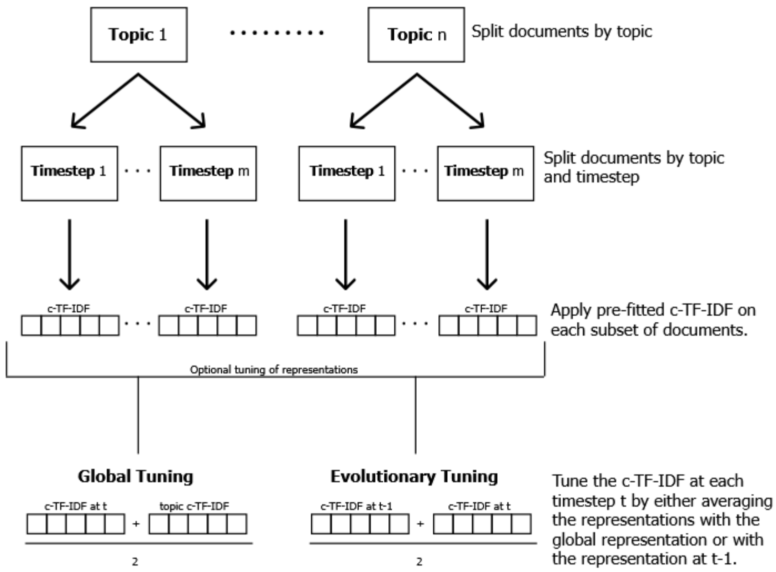


Figure 9: The BERTopic pipeline for dynamic topic modelling (Maarten Grootendorst, 2024)

For our use-case, we chose to embed our words using the French camemBERT model (Martin et al., 2020), suited to most of the language used in the considered documents. We also specify the count vectorizer method to again, get rid of accentuation, and keep only relevant words just as before.

First, we fit this BERTopic model over all political groups, to see if the spot of ecology subjects evolved over time in the global public debate. We decided to consider 20 topics this time, for a more nuanced approach. We can then plot the relative occurrence of the considered topic according to time. We notice that two topics seem closely related to ecology, topic 11 and topic 17 10.

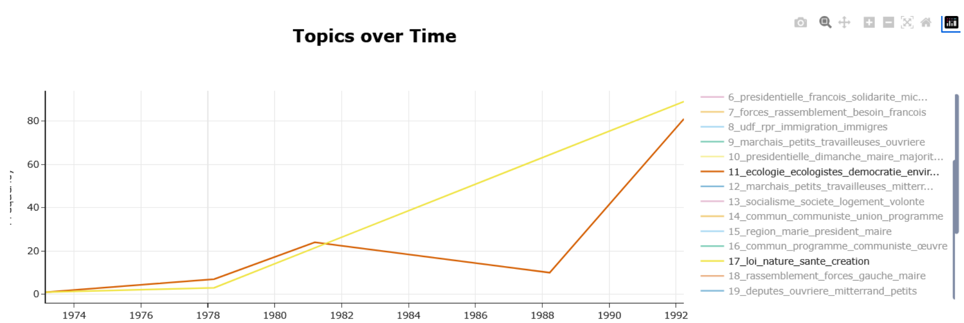


Figure 10: Evolution of the occurrence of topics 11 and 17 across time

We can see that those topic have a huge raise of popularity since the 70's, which shows that more and more parties started to take those matters into consideration.

Another thing we can notice is that the top words for topic 0 11, which seems closely related to communism and far-left ideas, change over time to finally include the word "environnement" in 1993.

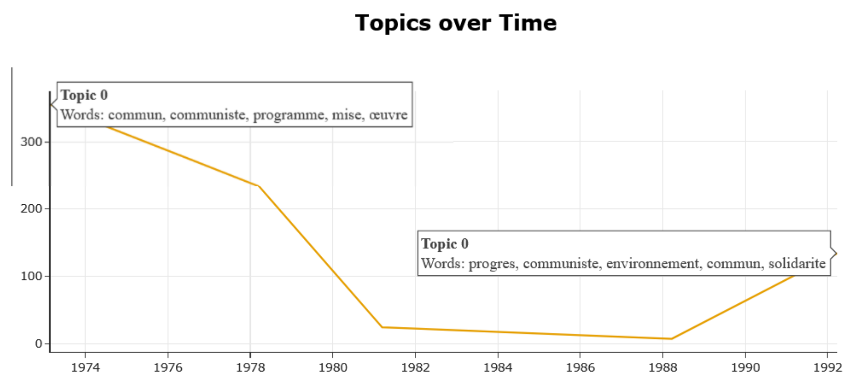


Figure 11: Difference in the most frequent words in topic 0 between 1973 and 1993

We can again established a link between ecology and left themes. Once again, we want to investigate those ties, by looking closer at both the far-left topic evolutions; and those of the ecologist political family.

It is thus of interest to us to see if the far-left group indeed had this topic among the most prevalent subjects of their campaigns toward the end of the century. We fit a BERTopic this time for 5 topics for the far-left political group 12, to check this assumption.

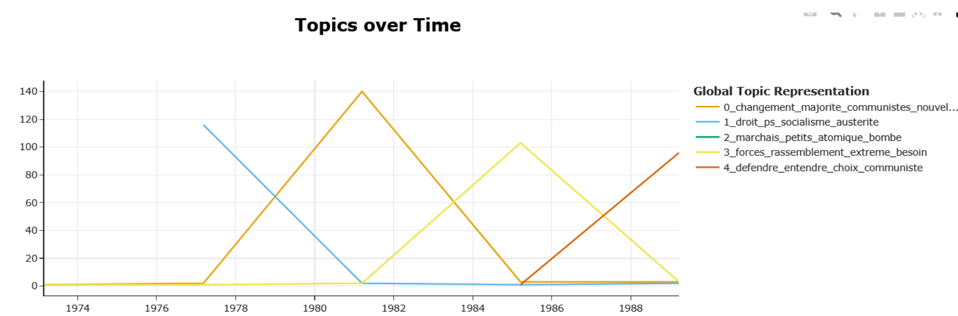


Figure 12: Dynamic topic modelling for the far-left political group

We can indeed see that all those topic occurrences vary greatly over time, as different historic phenomena take place, however we do not notice any mention of environmental matters in the popular topics.

Finally, we want to examine how the ecologist group's topic are changing overtime. After fitting a BERTopic on the ecologist group considering at most 5 topic 13, we also notice that only one topic is really worth to consider, and that the word "left" is in the top words, showing again that ecologist parties relate to the the political left.

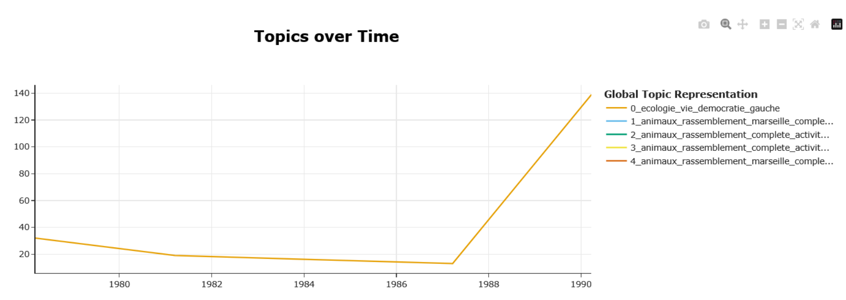


Figure 13: Dynamic topic modelling for the Ecologist group

6 Conclusion

The first finding of this work is the confirmation that ecology is a very recent topic in the general public debate, even if the ecologist parties have been advocating for the environment for a long time.

We also see that the ecologist parties are deeply ingrained in the political left, but the opposite isn't necessarily true, as the left and far-left have been neglecting those subjects.

Finally, we notice that those themes are on the rise, both in the global political landscape and in the ecologist parties, but since the other specific group don't denote of the same trends, we can also guess that it's a testimony of the rise of the number of ecologist candidate itself.

References

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A Variables available

This appendix describes the variables available for each manifesto in the dataset. Each row corresponds to one electoral document associated with a candidate in a given constituency and election round. The variables include document metadata, electoral context, geographical information, and biographical or political information about the main candidate and the substitute candidate.

Table 2: Document and election metadata variables.

Variable	Description
`id`	Unique identifier of the manifesto document.
`date`	Date associated with the election or electoral round.
`subject`	General subject tags describing the document, such as election type, country, and institutional context.
`title`	Full title of the document, usually including the election year, department, constituency, candidate name, and round.
`contexte-election`	Type of election, for example legislative election or partial legislative election.
`contexte-tour`	Election round associated with the manifesto.
`cote`	Archival or collection reference assigned to the document.
`images`	URL or list of URLs pointing to the scanned images of the manifesto.
`pdf`	URL pointing to the PDF version of the manifesto.
`ocr_url`	URL pointing to the OCR text extracted from the manifesto.

Table 3: Geographical and constituency variables. These variables locate each manifesto within the French electoral geography.

Variable	Description
`departement`	Department code associated with the constituency.
`departement-nom`	Name of the French department.
`departement-insee`	INSEE department code and department name.
`identifiant de circonscription`	Constituency identifier within the department.

Table 4: Main candidate variables. These variables describe the main candidate associated with each manifesto.

Variable	Description
`titulaire-nom`	Last name of the main candidate.
`titulaire-prenom`	First name of the main candidate.
`titulaire-sexe`	Gender of the main candidate.
`titulaire-age`	Age or year of birth as mentioned in the document.
`titulaire-age-calcule`	Calculated age of the main candidate.
`titulaire-age-tranche`	Age group of the main candidate.
`titulaire-profession`	Profession or occupational status of the main candidate.
`titulaire-mandat-en-cours`	Current political mandates held by the main candidate.
`titulaire-mandat-passe`	Former political mandates held by the main candidate.
`titulaire-associations`	Associations or organizational affiliations mentioned for the main candidate.
`titulaire-autres-statuts`	Other statuses mentioned for the main candidate, such as veteran, resistant, retired, or similar categories.
`titulaire-soutien`	Political support or party endorsement of the main candidate.
`titulaire-liste`	Electoral list or political list associated with the main candidate.
`titulaire-decorations`	Decorations or honors mentioned for the main candidate.

Table 5: Substitute candidate variables. These variables describe the substitute candidate associated with each manifesto.

Variable	Description
`suppleant-nom`	Last name of the substitute candidate.
`suppleant-prenom`	First name of the substitute candidate.
`suppleant-sexe`	Gender of the substitute candidate.
`suppleant-age`	Age or year of birth as mentioned for the substitute candidate.
`suppleant-age-calcule`	Calculated age of the substitute candidate.
`suppleant-age-tranche`	Age group of the substitute candidate.
`suppleant-profession`	Profession or occupational status of the substitute candidate.
`suppleant-mandat-en-cours`	Current political mandates held by the substitute candidate.
`suppleant-mandat-passe`	Former political mandates held by the substitute candidate.
`suppleant-associations`	Associations or organizational affiliations mentioned for the substitute candidate.
`suppleant-autres-statuts`	Other statuses mentioned for the substitute candidate.
`suppleant-soutien`	Political support or party endorsement of the substitute candidate.
`suppleant-liste`	Electoral list or political list associated with the substitute candidate.
`suppleant-decorations`	Decorations or honors mentioned for the substitute candidate.

B Top 50 words and pairs of words

Table 6: Most frequent unigrams and bigrams in the corpus. The two tables report the top 50 words and top 50 word pairs with their number of occurrences.

(a) Top 50 words

Word	Count
gauche	39968
france	37665
travailleurs	30682
français	29498
vie	26956
majorité	25580
faut	25236
pays	24768
travail	21364
nationale	20352
socialiste	20307
suppléant	20238
être	19602
programme	19504
député	19421
communiste	19372
pouvoir	18951
gouvernement	18595
tour	16747
sociale	16590
national	16154
conseiller	15684
république	15514
chômage	15212
maire	15208
mitterrand	15095
confiance	14592
société	14522
été	14473
même	14175
changement	14160
hommes	14086
nouvelle	14018
femmes	13742
voix	13720
président	13694
l'union	13502
général	13484
lutte	13342
commun	13330
progrès	13114
liberté	12783
enfants	12192
jeunes	12068
droit	12031
économique	12027
française	11899
qu'ils	11875
entreprises	11864
j'ai	11695

(b) Top 50 bigrams

Bigram	Count
programme commun	9620
l'assemblée nationale	7907
front national	7205
conseiller général	6720
françois mitterrand	6637
communiste français	5199
république française	5044
conseiller municipal	4861
président république	4753
électrices électeurs	4159
député sortant	3799
l'union gauche	3761
lutte ouvrière	3669
justice sociale	3546
marié enfants	3538
majorité présidentielle	3447
travailleuses travailleurs	3316
sécurité sociale	3248
dès tour	3218
hommes femmes	3152
nouvelle majorité	3131
petits commerçants	2912
pouvoir d'achat	2900
progrès social	2883
rpr l'udf	2837
electrices electeurs	2745
gouvernement gauche	2617
commerçants artisans	2599
conseil général	2564
second tour	2563
radicaux gauche	2495
france unie	2484
hommes politiques	2428
lutte ouvriere	2311
conseiller régional	2254
changer vie	2144
cadre vie	2134
services publics	2071
adjoint maire	2059
forces gauche	2057
peine mort	2027
giscard d'estaing	2024
dimanche prochain	2008
d'union gauche	1917
allocations familiales	1893
qualité vie	1832
protection sociale	1772
commun gouvernement	1731
même temps	1729
ministres communistes	1723